Laborator 4

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Grupă : 321CC

Metoda Master

↳ caz 2 → caz 1 → caz 3

1. Rezolvați folosind metoda Master :

a) $T(n) = 16 \cdot T(n/4) + n^2$

$T(n) = a\,T(n/b) + f(n)$

Caz 2: Dacă $f(n) = \Theta(n^{\log_b^a})$ atunci $T(n) = \Theta(n^{\log_b^a} \cdot \lg n)$

$a = 16$, $b = 4$ $\Rightarrow$ $n^{\log_b^a} = n^{\log_4^{16}} = n^{\log_4^{4^2}} = n^2$,

dar $f(n) = n^2 \Rightarrow$

$f(n) = \Theta(n^2) \Rightarrow$ conform cazului 2 din teorema Master $\Rightarrow$

$$T(n) = \Theta(n^{\log_4^{16}} \cdot \lg n) = \Theta(n^2 \cdot \lg n)$$

b) $T(n) = 4 \cdot T(n/2) + \Theta(n)$

$T(n) = a\,T(n/b) + f(n)$

Caz 2: Dacă $f(n) = \Theta(n^{\log_b^a})$ atunci $T(n) = \Theta(n^{\log_b^a} \cdot \lg n)$

$\left.\begin{matrix} a=4 \\ b=2 \end{matrix}\right\} \Rightarrow n^{\log_b^a} = n^{\log_2^4} = n^{\log_2^{2^2}} = n^2$

$f(n) = \Theta(n) = kn,\ k \in \mathbb{R}$ constantă

Cum $kn \neq \Theta(n^2) \Rightarrow$ nu putem aplica cazul 2 al teoremei Master

$T(n) = a\,T(n/b) + f(n)$

Caz 1: Dacă $\exists\, \varepsilon > 0$ astfel încât $f(n) = O(n^{\log_b^a - \varepsilon})$ atunci $T(n) = \Theta(n^{\log_b^a})$

$\left.\begin{matrix} a=4 \\ b=2 \end{matrix}\right| \Rightarrow n^{\log_b^a} = n^{\log_2^{2^2}} = n^2$

$f(n) = \Theta(n) = \Theta(n^1) = \Theta(n^{2-1}) =$

$\Theta(n^{\log_b^a - 1}) \Rightarrow f(n) = O(n^{\log_b^a - 1}) \Rightarrow$

$\exists\, \varepsilon = 1 > 0 \Rightarrow$ aplicăm cazul 1 al a.î. $f(n) = O(n^{\log_b^a - \varepsilon})$

teoremei master $\Rightarrow T(n) = \Theta(n^{\log_b^a}) = \Theta(n^2)$

c) $T(n) = 2 \cdot T(n/2) + \Theta(n^3)$

$T(n) = a\,T(n/b) + f(n)$

Caz 2: Dacă $f(n) = \Theta(n^{\log_b^a})$ atunci $T(n) = \Theta(n^{\log_b^a} \cdot \lg n)$

$a = 2$, $b = 2$ $\Rightarrow$ $n^{\log_b^a} = n^{\log_2^2} = n^1$

$f(n) = \Theta(n^3)$

Cum $n^1 \neq \Theta(n^3) \Rightarrow$ nu putem aplica cazul 2 al teoremei master

$T(n) = a\,T(n/b) + f(n)$

Caz 1: Dacă $\exists \varepsilon > 0$ astfel încât $f(n) = O(n^{\log_b^a - \varepsilon})$ atunci $T(n) = \Theta(n^{\log_b^a})$

$f(n) = \Theta(n^3)$

$a = 2$, $b = 2$ $\Rightarrow$ $n^{\log_b^a} = n^{\log_2^2} = n$

Cum nu există $\varepsilon > 0$ astfel încât

$k \cdot n^3 = O(n^{1-\varepsilon})$, $k \in \mathbb{R}$ constantă $\Rightarrow$

nu putem aplica cazul 1 al teoremei Master.

$T(n) = aT(n/b) + f(n)$

Caz 3: Dacă $\exists \varepsilon > 0$ a.î. $f(n) = \Omega(n^{\log_b^a + \varepsilon})$

și $\exists c \in (0,1)$, $n_0 \in \mathbb{N}^*$ a.î. $af(n/b) \leq cf(n)$,

$(\forall) n \geq n_0$ atunci $T(n) = \Theta(f(n))$

$a=2$
$b=2$ $\Big| \Rightarrow$ $n^{\log_b^a} = n^{\log_2^2} = n$

$f(n) = \Theta(n^3) = k \cdot n^3$, $k \in \mathbb{R}$ constantă

i) $\exists \varepsilon > 0$ a.î. $f(n) = \Omega(n^{\log_b^a + \varepsilon})$

Dem.

$\exists c' \in \mathbb{R}_+^*$, $\exists n_0' \in \mathbb{N}^*$ a.î.

$c' \cdot n^{\log_b^a + \varepsilon} \leq f(n)$, $(\forall) n \geq n_0' \Rightarrow$

$c' \cdot n^{\varepsilon} \leq n^3$, $(\forall) n \geq n_0'$, adevărat

pentru $\varepsilon = 2$, $n_0 = 1$, $c' = 1 \Rightarrow$

$\exists \varepsilon > 0$ a.î. $f(n) = \Omega(n^{\log_b^a + \varepsilon})$

ii) $\exists c \in (0,1)$, $\exists n_0 \in \mathbb{N}^*$ a.î.

Dem.

$a f(n/b) \leq c f(n)$, $(\forall) n \geq n_0$

$2 f(n/2) \leq c f(n)$, $(\forall) n \geq n_0 \Rightarrow$

$2k \left(\frac{n}{2}\right)^3 \leq c \cdot k n^3$, $(\forall) n \geq n_0 \Rightarrow$

$\frac{k}{4} \cdot n^3 \leq c \cdot k n^3$, $(\forall) n \geq n_0$

Pentru $n_0 = 1$ obținem:

$\frac{1}{4} \leq c$

Pentru $c = \frac{1}{2} \in (0,1)$, inegalitatea este adevărată

$\Rightarrow \exists c \in (0,1)$, $\exists n_0 \in \mathbb{N}^*$ a.î.

$a f(n/b) \leq c f(n)$, $(\forall) n \geq n_0$

Conform i) și ii) al cazului 3 al metodei master $\Rightarrow T(n) = \Theta(n^3)$

d) $T(n) = 2 \cdot T(n/2) + \Theta(n \lg(n))$

$a=2$, $b=2$ $\Rightarrow$ $n^{\log_b^a} = n^1 = n$

$f(n) = \Theta(n \lg(n))$

$f(n) = k \cdot n \lg(n)$, unde $k \in \mathbb{R}$ constantă

$T(n) = a\,T(n/b) + f(n)$

Caz 2: Dacă $f(n) = \Theta(n^{\log_b^a})$ atunci $T(n) = \Theta(n^{\log_b^a} \cdot \lg n)$

$\Theta(n^{\log_b^a}) = \Theta(n)$

$\Theta(n) = k n \lg(n)$, fals $\Rightarrow$ nu se poate aplica cazul 2 al teoremei Master

$T(n) = a\,T(n/b) + f(n)$

Caz 1: Dacă $\exists \varepsilon > 0$ astfel încât $f(n) = O(n^{\log_b^a - \varepsilon})$ atunci $T(n) = \Theta(n^{\log_b^a})$

$a=2$, $b=2$ $\Rightarrow$ $n^{\log_b^a} = n$

$f(n) = kn \lg(n)$, $k \in \mathbb{R}$

$\exists \varepsilon > 0$ astfel încât $n \lg n = O(n^{1-\varepsilon})$,

fals $\Rightarrow$ nu putem folosi cazul 1
al teoremei master

$T(n) = a\,T(n/b) + f(n)$

Caz 3: Dacă $\exists \varepsilon > 0$ aî. $f(n) = \Omega(n^{\log_b a + \varepsilon})$

și $\exists c \in (0,1)$, $n_0 \in \mathbb{N}^*$ a.î. $a f(n/b) \le c f(n)$,

$(\forall)\ n \ge n_0$ atunci $T(n) = \Theta(f(n))$

Verificăm dacă $\exists\ \varepsilon > 0$ astfel încât

$f(n) = \Omega(n^{1+\varepsilon})$

$f(n) = k n \lg n$, $k \in \mathbb{R}_+^*$ constantă

$\exists c' \in \mathbb{R}_+^*$, $\exists n_0 \in \mathbb{N}^*$ a.î.

$c' \cdot n^{1+\varepsilon} \le k n \lg n$, $(\forall) n \ge n_0 \Rightarrow$

$c' \cdot n^{\varepsilon} \le k \lg n$, $(\forall) n \ge n_0 \Rightarrow$

$\frac{c'}{k} \le \frac{\lg n}{n^{\varepsilon}} \xrightarrow{n \to +\infty} 0$, fals deoarece

$c' \in \mathbb{R}_+^*$, $k \in \mathbb{R}_+^* \Rightarrow$

Nu se poate aplica teorema Master

2. a) $T(n) = 2 \cdot T(n/2) + \Theta(n \lg n)$

$f(n) = \Theta(n \lg n)$

$f(n) = h \cdot n \lg n$, $h \in \mathbb{R}_+^*$ constantă

Vom folosi metoda iterativă

$T(n) = 2T(n/2) + hn \lg n \quad | \cdot 2^0$

$T(n/2) = 2T(n/2^2) + h\frac{n}{2} \lg \frac{n}{2} \quad | \cdot 2^1$

$\vdots$

$T(n/2^k) = 2T(n/2^{k+1}) + h\frac{n}{2^k} \lg \frac{n}{2^k} \quad | \cdot 2^k$

unde $2^{k+1} = n$, $k+1 = \lg n$

(+)

$$T(n) = 2^{k+1} \cdot T(n/2^{k+1}) + \sum_{i=0}^{k} h \cdot n \cdot \lg \frac{n}{2^i} =$$

$$= n \cdot T(1) + \sum_{i=0}^{k} h \cdot n \lg n - \sum_{i=0}^{k} h \cdot n \lg 2^i$$

$$= nT(1) + hnk \lg n - hn \cdot \frac{k(k+1)}{2} =$$

$$= nT(1) + hn(\lg n)^2 - hn \frac{\lg n (\lg n + 1)}{2}$$

$$= nT(1) + hn(\lg n)^2 - \frac{1}{2} hn(\lg n)^2 -$$

$\frac{1}{2} n \lg n = n(\lg n)^2 - \frac{1}{2} n \lg n$

$+ n\Theta(1) \Rightarrow$

$T(n) = \Theta(n \lg^2 n)$

b) Rezolvați recurența

$T(n) = 2 \cdot T(n-1) + \Theta(1)$

Voi folosi metoda iterativă

$T(n) = 2T(n-1) + 1 \quad | \cdot 2^0$

$T(n-1) = 2T(n-2) + 1 \quad | \cdot 2^1$

$\vdots$

$T(n-k) = 2T(n-k-1) + 1 \quad | \cdot 2^k$

unde $n-k-1 = 1 \Rightarrow k+1 = n-1$

(+)

$T(n) = 2^{k+1} \cdot T(n-k-1) + \sum_{i=0}^{k} 2^i =$

$= 2^{k+1} \cdot T(1) + (2^{k+1} - 1) =$

$= 2^{n-1} \cdot T(1) + (2^{n-1} - 1) =$

$= 2^{n-1} \cdot \Theta(1) + 2^{n-1} - 1 = 2^n - 1 = \Theta(2^n - 1)$